TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of Available Chlorine in Bleaching Powder by the Iodometric Method

Chlorine measured without ever handling chlorine — the powder's oxidising power first liberated as iodine, then the iodine counted drop by drop with thiosulphate and starch.

CHE-1206 · CHEMISTRY-II SESSIONAL · EXPERIMENT 11 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to determine the percentage of **available chlorine** in a sample of bleaching powder. Bleaching powder, approximately Ca(OCl)Cl, bleaches and disinfects through its hypochlorite. Its quality is not stated as a hypochlorite percentage but as **available chlorine**: the mass of chlorine gas that has the *same oxidising power* as a given mass of the powder — because on acidification the powder itself releases exactly that oxidising power as chlorine:

$$OCl^- + Cl^- + 2\ H^+ \rightarrow Cl_2 + H_2O$$

Chlorine is a poisonous gas and a miserable thing to collect, so the determination is **iodometric** — the oxidising power is transferred to iodine, which is easy to titrate. The powder is shaken with water, and an aliquot of the suspension is treated with excess potassium iodide and acetic acid; the hypochlorite liberates an exactly equivalent amount of iodine:

$$OCl^- + 2\ I^- + 2\ H^+ \rightarrow I_2 + Cl^- + H_2O \quad \text{and, in parallel,} \quad Cl_2 + 2\ I^- \rightarrow I_2 + 2\ Cl^-$$

The liberated iodine is titrated with standard sodium thiosulphate, starch serving as indicator near the end:

$$I_2 + 2\ S_2O_3^{2-} \rightarrow 2\ I^- + S_4O_6^{2-}$$

The equivalence chain is the whole calculation: **1 mol Cl_2 ≡ 1 mol I_2 ≡ 2 mol $S_2O_3^{2-}$**, so 1 mL of 0.100 M thiosulphate corresponds to 3.545 mg of chlorine. Acetic acid, not a strong mineral acid, sets the pH: strong acid would decompose the thiosulphate during the titration and would speed up the air-oxidation of iodide, inventing iodine that the powder never made.

2. Apparatus and solutions

- Standard sodium thiosulphate solution, 0.100 M, in the burette.
- Bleaching powder sample, 2.50 g made up to 250 mL with water (a suspension — the powder only partly dissolves).
- Potassium iodide solution (10 %) in excess; glacial/dilute acetic acid; starch solution (freshly prepared).
- Pipette 25 mL, conical flask, mortar and pestle, white tile.

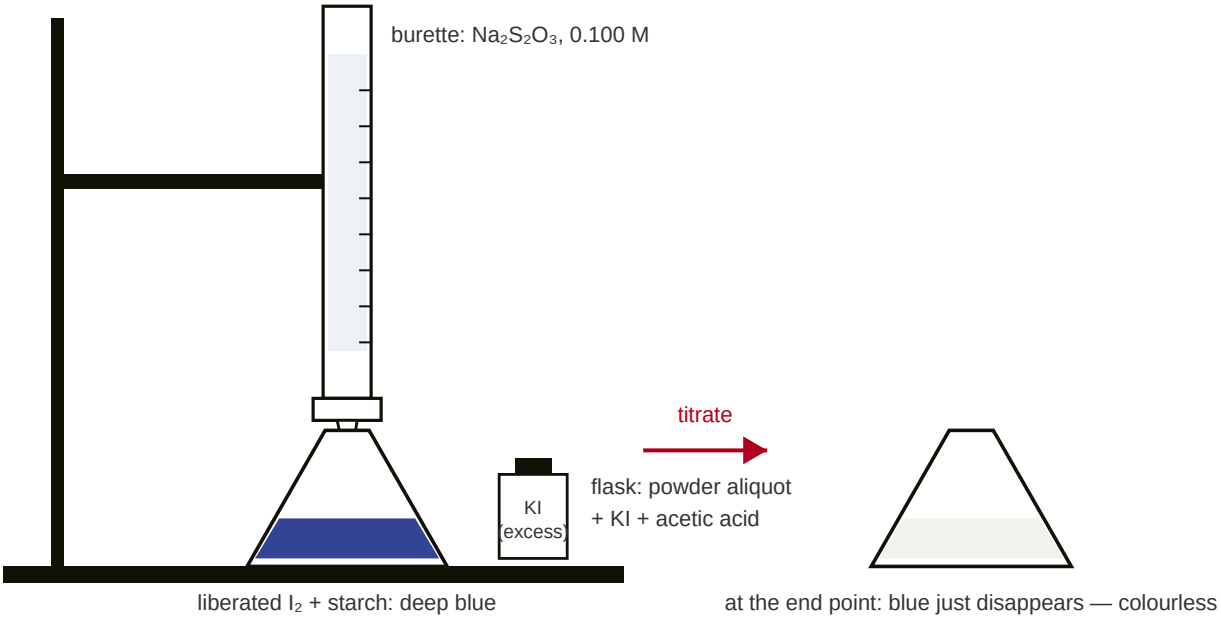


Figure 1 — The iodometric titration. Starch turns the liberated iodine deep blue; thiosulphate removes the iodine, and the end point is the drop that discharges the blue completely. Starch is added only when the iodine colour has faded to pale straw (see Procedure).

3. Procedure

1. Weigh 2.50 g of bleaching powder, grind it in a mortar with a little water to a smooth paste, and wash it quantitatively into a 250 mL volumetric flask; make up to the mark and shake well. The liquid is a suspension — that is expected.
2. Rinse and fill the burette with the standard thiosulphate.
3. Shake the flask, and **immediately** pipette 25.0 mL of the suspension (≡ 0.250 g of powder) into a conical flask, before the solid settles.
4. Add about 10 mL of potassium iodide solution (a clear excess) and about 10 mL of dilute acetic acid. Iodine is liberated at once — the solution turns deep brown.

5. Titrate with thiosulphate, swirling, until the brown fades to a **pale straw yellow**. Only now add 1–2 mL of starch solution: the contents turn deep blue.
6. Continue dropwise until one drop discharges the blue completely — the solution becomes **colourless**. Record the titre, and work promptly throughout: liberated iodine is volatile, and acidified iodide is slowly oxidised by air.
7. Repeat until two titres agree within 0.1 mL; average the concordant titres.

4. Worked calculation (model data)

AVAILABLE-CHLORINE RESULT

Sample taken in the aliquot: 0.250 g. Standard thiosulphate: 0.100 M. Concordant titre: 25.4 mL.
Moles of $S_2O_3^{2-}$ = 0.100 × 0.0254 = 2.54×10^{-3} mol.
Moles of I_2 = moles of Cl_2 equivalent = $2.54 \times 10^{-3} / 2 = 1.27 \times 10^{-3}$ mol (chain: $Cl_2 \equiv I_2 \equiv 2\ S_2O_3^{2-}$).
Mass of available chlorine = $1.27 \times 10^{-3} \times 70.9 = 0.0900$ g, in 0.250 g of powder.
Available chlorine = 0.0900/0.250 × 100 = **36.0 %**.
Bench rule (derive it, don't just use it): 1 mL of 0.100 M $S_2O_3^{2-} \equiv 3.545$ mg Cl — check: 25.4 × 3.545 = 90.0 mg ✓. Fresh commercial bleaching powder carries roughly 35–38 % available chlorine, so the result is in the expected range; an old, air-exposed sample reads far lower.

5. Observation table (as recorded)

TRIAL	INITIAL READING (ML)	FINAL READING (ML)	TITRE (ML)
1 (rough)	0.0	25.9	25.9
2	0.0	25.4	25.4
3	0.0	25.4	25.4

Note the two-stage colour journey of every run: brown (much iodine) → straw (little iodine) → deep blue (starch added) → colourless (none). Each stage has its own instruction, which is why the rough run matters.

6. Precautions

- Shake the suspension and pipette at once: the active solid settles in seconds, and a pipette drawn from a settled flask samples water, not powder (result low and irreproducible).

- Add starch only at the pale-straw stage. Added early, the starch–iodine complex is so strong and colloidal that iodine is released back only sluggishly, and the end point drags past the truth.
- Titrate immediately after acidifying: iodine evaporates from an open flask, and acidified iodide left standing is oxidised by air to extra iodine — the two errors pull opposite ways, and prompt work shrinks both.
- Use acetic acid, not hydrochloric or sulphuric: strong acid decomposes thiosulphate ($S_2O_3^{2-} + 2\ H^+ \rightarrow S\downarrow + SO_2 + H_2O$) and accelerates air-oxidation of iodide.
- Keep the iodide in clear excess — any hypochlorite that finds no I^- escapes as uncounted chlorine gas.
- Store the powder sample stoppered and assay it fresh: bleaching powder loses available chlorine to air and moisture day by day, which is itself one of the facts this experiment exists to show.

7. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Define ‘available chlorine’ precisely, and explain why bleaching powder is graded by it rather than by its hypochlorite content.
2. Why is acetic acid used to acidify, instead of dilute hydrochloric acid?
3. Why is the starch withheld until the solution is pale straw-yellow?
4. An aliquot representing 0.250 g of powder needs 21.6 mL of 0.100 M thiosulphate. Find the available chlorine percentage.
5. Why must the titration follow the acidification immediately? Name the two opposing errors that grow with delay.
6. Why does a tin of bleaching powder left open in the laboratory assay lower every month, even though nothing is taken out of it?

Answer key

1. Available chlorine is the mass of chlorine gas having the same oxidising power as the sample — equivalently, the chlorine the sample would release on acidification. It grades the powder by what it can *do* (bleach, disinfect, oxidise) in chlorine units every user can compare, whatever the powder’s exact and variable composition.
2. Strong acid attacks the titrant itself — thiosulphate decomposes to sulphur and sulphur dioxide in strongly acid solution — and it speeds the air-oxidation of I^- to I_2, which the titration would credit to the powder. Acetic acid is strong enough to liberate the iodine quantitatively (the hypochlorite reaction needs only mild acidity) without either side reaction.

3. At high iodine concentration the starch forms a dense, colloidal complex that gives its iodine back very slowly; the blue would linger past the true end point and the titre would run high. At pale-straw concentration the complex forms and breaks cleanly, drop by drop.
4. Chlorine = 21.6 × 3.545 mg = 76.6 mg in 0.250 g → 76.6/250 × 100 = 30.6 % available chlorine — a sample that has lost strength on storage.
5. Liberated iodine is volatile and escapes from solution (titre falls — result low), while dissolved air oxidises the abundant iodide in acid solution to fresh iodine (titre rises — result high). Immediate titration keeps both small instead of trusting them to cancel.
6. The powder reacts with moist air and its CO_2: hypochlorite is converted and chlorine is slowly lost ($Ca(OCl)Cl$ degrades toward $CaCO_3$, $CaCl_2$ and chlorate). Its oxidising power — the thing being assayed — literally leaves as gas and reaction products, so the percentage falls although the tin's contents were never touched.

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